

Since $|g(x)| \leq x^2$ for all x , plugging $x=0$ gives $|g(0)| \leq 0$ so $g(0)=0$ using derivative definition:

$$g'(0) = \lim_{h \rightarrow 0} (g(h) - g(0))/h = \lim_{h \rightarrow 0} g(h)/h$$

Since $|g(h)| \leq h^2$, we get: $|g(h)/h| \leq |h|$ and $|h| \rightarrow 0$ as $h \rightarrow 0$ so $g'(0) = 0$ so $g(0)=0; g'(0)=0$

1(b) (i) $\rightarrow g(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $g(0)=0$

differentiability at 0: $g'(0) = \lim_{x \rightarrow 0} (g(x) - g(0))/x$
 $= \lim_{x \rightarrow 0} x \sin(1/x)$ since $|x \sin(1/x)| \leq |x|$ and $|x| \rightarrow 0$, limit is 0 so g is differentiable at 0 and $g'(0)=0$

1(b) (ii) $h(x) = x \cos(1/x)$ for $x \neq 0$ and $h(0)=0$

then $h'(0) = \lim_{x \rightarrow 0} (h(x) - h(0))/x = \lim_{x \rightarrow 0} \cos(1/x)$
 So this limit doesn't exist because $\cos(1/x)$ oscillates
 $\rightarrow h$ not differentiable at 0

$$2(i) \lim_{x \rightarrow 0} \frac{\sin(x^2)}{\sin^2(3x)}$$

$$\frac{\sin(x^2)}{\sin^2(3x)} = \frac{\sin(x^2)}{x^2} \cdot \frac{x^2}{\sin^2(3x)} = \frac{\sin(x^2)}{x^2} \cdot \left(\frac{x}{\sin(3x)} \right)^2$$

$$\frac{x}{\sin 3x} = \frac{1}{3} \cdot \frac{3x}{\sin 3x} \quad \text{so} \quad \lim_{x \rightarrow 0} \frac{\sin x^2}{\sin^2(3x)} =$$

$$\left(\lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} \right) \left(\lim_{x \rightarrow 0} \frac{x}{\sin(3x)} \right)^2 = 1 \cdot \left(\frac{1}{3} \right)^2 = \boxed{\frac{1}{9}}$$

$$2(ii) \lim_{x \rightarrow 1} \frac{(1 - \cos(\pi x)) \sin(x-1)}{x^2 - 1}$$

$$\text{So } x^2 - 1 = (x-1)(x+1) \text{ then } \frac{(1 - \cos(\pi x)) \sin(x-1)}{x^2 - 1} \\ = \frac{1 - \cos(\pi x)}{x+1} \cdot \frac{\sin(x-1)}{x-1}$$

As $x \rightarrow 1$

$$1 - \cos(\pi x) \rightarrow 1 - \cos(\pi) = 2$$

$$x+1 \rightarrow 2$$

$$\text{and } \frac{\sin(x-1)}{x-1} \rightarrow 1 \text{ So } \lim_{x \rightarrow 1} \frac{(1 - \cos(\pi x)) \sin(x-1)}{x^2 - 1} \\ = \frac{2}{2} \cdot 1 = 1$$

$$3(a) \ y = a \sin x + b \cos x \text{ so } y' = a \cos x - b \sin x$$

$$y'' = -a \sin x - b \cos x \text{ then } 3y'' - y' + 2y = 3 \sin x + 2 \cos x$$

$$3(-a \sin x - b \cos x) - (a \cos x - b \sin x) + 2(a \sin x + b \cos x)$$

$$\text{So } (-a+b) \sin x + (-a-b) \cos x \text{ So } -a+b=1$$

$$\text{Since } -2a=3 \text{ and } a=-\frac{3}{2} \text{ So } -a-b=2$$

$$\Rightarrow \frac{3}{2} + b = 1 \Rightarrow b = -\frac{1}{2} \text{ So } a = -\frac{3}{2}, b = -\frac{1}{2}$$

$$3(b) \ e^{2x} \cos t = xt + 2e \text{ so } \frac{d}{dt} (e^{2x} \cos t) = \frac{d}{dt} (xt + 2e)$$

$$e^{2x} (2x' \cos t - \sin t) = x' + t \text{ So } 2e^{2x} x' \cos t - e^{2x} \sin t = x' + t$$

$$\text{and } x' (2e^{2x} \cos t - t) = x + e^{2x} \sin t \text{ so } \frac{dx}{dt} = \frac{x + e^{2x} \sin t}{2e^{2x} \cos t - t}$$

$$\text{at } t=0 \text{ we get } e^{2x} = 2e \text{ so } x(0) = 1 + \ln 2$$

$$x'(0) = \frac{x(0) + e^{2x(0)} \sin 0}{2e^{2x(0)} \cos 0 - 0} = \frac{x(0)}{2e^{2x(0)}}$$

since ~~then~~ $e^{2x(0)} = 2e$ then $x'(0) = \frac{x(0)}{4e} = \frac{1 + \ln 2}{8e}$
 so $\frac{dx}{dt} = \frac{x + e^{2x} \sin t}{2e^{2x} \cos t - 1}$, $x(0) = 1 + \ln 2$, $x'(0) = \frac{1 + \ln 2}{8e}$

4) we got $e^{\cos x} = c$ since $-1 \leq \cos x \leq 1$
 so $e^{-1} \leq e^{\cos x} \leq e$ so possible values are $\frac{1}{e} \leq c \leq e$

4) ii) $f'(x) = e^{\cos x} (-\sin x) = -e^{\cos x} \sin x$

$f''(x) = -[-e^{\cos x} \sin x] \sin x + e^{\cos x} \cos x = e^{\cos x} \sin^2 x - e^{\cos x} \cos x$ so $f''(x) = e^{\cos x} (\sin^2 x - \cos x)$

4) iii) $f''(x) = e^{\cos x} (\sin^2 x - \cos x)$ since $e^{\cos x} > 0$ for all x then $\sin^2 x - \cos x = 0$ for some $x \in (\frac{\pi}{4}, \frac{\pi}{3})$
 $g(x) = \sin^2 x - \cos x$ is continuous so:

$g(\frac{\pi}{4}) = \frac{1}{2} - \frac{\sqrt{2}}{2} < 0$ and $g(\frac{\pi}{3}) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4} > 0$

since g is continuous and changes sign on $(\frac{\pi}{4}, \frac{\pi}{3})$
 The intermediate value theorem says that for $x \in (\frac{\pi}{4}, \frac{\pi}{3})$

where $g(x) = 0$ so $f''(x) = 0$ for x between $\frac{\pi}{4}$ and $\frac{\pi}{3}$
 graph) $\rightarrow f'(-2) = 0$, $f'(0) = 0$ and $f'(1) = -1$
 b) limit doesn't exist as a finite number